

DATA ANALYSIS PROJECT REPORT

Project Title	Sales Analysis: Customer Purchase Behavior & Insights
Student Name	MD. ABDUR RAUF
Student ID	23-50713-1
Course / Section	PROGRAMMING IN PYTHON [C]
Instructor	MD. TANZEEM RAHAT
Date of Submission	02-MAY-2026

1. Executive Summary

This project presents an exploratory data analysis (EDA) of a sales dataset containing transactional records across customer demographics, geographic regions, occupations, and product categories in India. The primary goal was to uncover key patterns in consumer purchasing behavior during the festive season to support data-driven marketing and inventory decisions. The analysis involved data cleaning including dropping irrelevant columns, handling null values, and correcting data types followed by grouped aggregations and visual exploration using Seaborn and Matplotlib. Three findings stood out most strongly: first, female customers consistently outpaced male buyers in both order volume and total spending; second, the 26–35 age group accounted for the largest share of purchases, with married women forming the single most active buyer segment; and third, Uttar Pradesh, Maharashtra, and Karnataka led all states in revenue, with Food, Clothing, and Electronics emerging as the top-selling product categories across IT, Healthcare, and Aviation professionals.

2. Introduction and Project Objectives

Millions of households across every income bracket are increasing their spending on food, clothing, gifts, and electronics. Understanding who is buying, what they are buying, and where they are buying from is therefore not just academically interesting; it carries real strategic value for retailers, e-commerce platforms, and marketers alike.

This analysis is motivated by the need to move beyond surface-level sales figures and into the behavioral patterns that drive them. By examining a rich transactional dataset spanning customer demographics, geographic regions, occupations, and product preferences, the goal is to identify the most valuable customer segments during the Diwali period. These insights can directly inform decisions around targeted advertising, regional inventory planning, and personalized promotions ultimately helping businesses maximize impact during India's most commercially significant season.

Item	Your Content
Dataset name	Sales Data.csv
Dataset source	https://www.kaggle.com/datasets/priyushwaha/diwali-sales-dataset
Main project goal	The insights derived aim to help businesses better understand their most valuable buyers and make informed decisions around targeted marketing, regional stocking, and festive season sales strategy.
Research Question 1	Who are the most active buyers in terms of gender, age group, and marital status, and how do these demographic factors influence total purchasing power?
Research Question 2	Which Indian states generate the highest sales volume, revenue and which product categories and occupational segments drive that demand?

Research Question 3	What is the profile of the highest-value customer, and how can this profile be used to guide targeted marketing and inventory decisions?
---------------------	--

3. Dataset Description

The dataset used in this project contains transactional and demographic records collected during the Diwali festive sales period in India. Each row represents a single customer transaction, capturing both the profile of the buyer and the details of their purchase. The dataset originally contained several columns, two of which `Status` and `unnamed1` were found to be irrelevant or entirely blank and were dropped during the cleaning phase. Null values were also removed, and the `Amount` column was converted from float to integer to ensure analytical consistency.

The variables available in the dataset span four broad categories:

- **Demographic:** Gender, Age, Age Group, Marital Status
- **Geographic:** State
- **Professional:** Occupation
- **Transactional:** Product ID, Product Category, Orders, Amount

The dataset represents buyers from across multiple Indian states, with a particular concentration of records from Uttar Pradesh, Maharashtra, and Karnataka. While an exact collection date is not specified, the transactional context clearly situates the data within India's Diwali festive season, making it a strong source for studying seasonal consumer behavior patterns.

3.1 Basic Dataset Information

Property	Value
Number of rows	11,251
Number of columns	15
Data types present	numerical, categorical, date-time, text
Target or key variable(s)	Amount, Orders, Gender, Age Group, etc.
Potential issues observed at first glance	missing values

3.2 Initial Observations

Upon first inspection of the Diwali sales dataset using `df.head()`, `df.info()`, and `df.describe()`, several immediate observations were made. Two columns `Status` and `unnamed1` — were found to be entirely blank or irrelevant and were removed early in the cleaning process, while the `Amount` column was stored as a float despite representing whole currency values, requiring a type conversion to integer. A small number of null values were detected across the dataset using `pd.isnull(df).sum()`, and the affected rows were dropped to ensure analytical integrity, reducing the dataset by approximately 12 records.

4. Data Cleaning and Preparation

Issue Found	Action Taken	Reason / Justification	Code Reference
Two irrelevant/blank columns (Status, unnamed1)	Dropped both columns using <code>df.drop(['Status', 'unnamed1'], axis=1, inplace=True)</code>	These columns contained no meaningful data and would have added noise to the analysis.	Data cleaning section cell 4
Null values detected across multiple columns	Identified using <code>pd.isnull(df).sum()</code> , then removed all affected rows using <code>df.dropna(inplace=True)</code>	Since the number of null rows was small (approximately 12 out of 11,251), dropping them was a safe and reasonable choice.	Data cleaning section cell 5
Amount column stored as incorrect data type (float)	Converted to integer using <code>df['Amount'].astype('int')</code>	Currency amounts in this context represent whole rupee values with no decimal component.	Data cleaning section cell 6
Non-standard character encoding in CSV file	Loaded the dataset with <code>encoding='unicode_escape'</code> inside <code>pd.read_csv()</code>	Indian datasets frequently contain regional language characters or special symbols that cause <code>UnicodeDecodeError</code> under default UTF-8 encoding.	Data import section — cell 1

5. Feature Engineering

New Feature	How It Was Created	Why It Is Useful
Age Group	Customers were segmented into labeled age brackets (e.g., 0–17, 18–25, 26–35, 36–45, 46–50, 51–55, 55+) from the raw Age numerical column	Continuous age values alone are difficult to visualize and compare. Grouping them into brackets reveals which life stage drives the most purchasing activity, directly answering Research Question 1

Total Sales by Group	Created using <code>df.groupby([variable], as_index=False)['Amount'].sum()</code> across Gender, State, Occupation, and Product Category	Created using <code>df.groupby([variable], as_index=False)['Amount'].sum()</code> across Gender, State, Occupation, and Product Category
Order Volume by Segment	Created using <code>df.groupby([variable], as_index=False)['Orders'].sum()</code> across State and Product ID	Separating order count from revenue amount allows a distinction between high-frequency low-value buyers and low-frequency high-value buyers an important nuance for inventory and marketing planning

6. Analysis Methodology

The analysis followed a structured exploratory data analysis (EDA) approach, progressing logically from data ingestion and cleaning through grouped aggregation and visual interpretation. Three core libraries formed the analytical backbone of the project, each serving a distinct and complementary role.

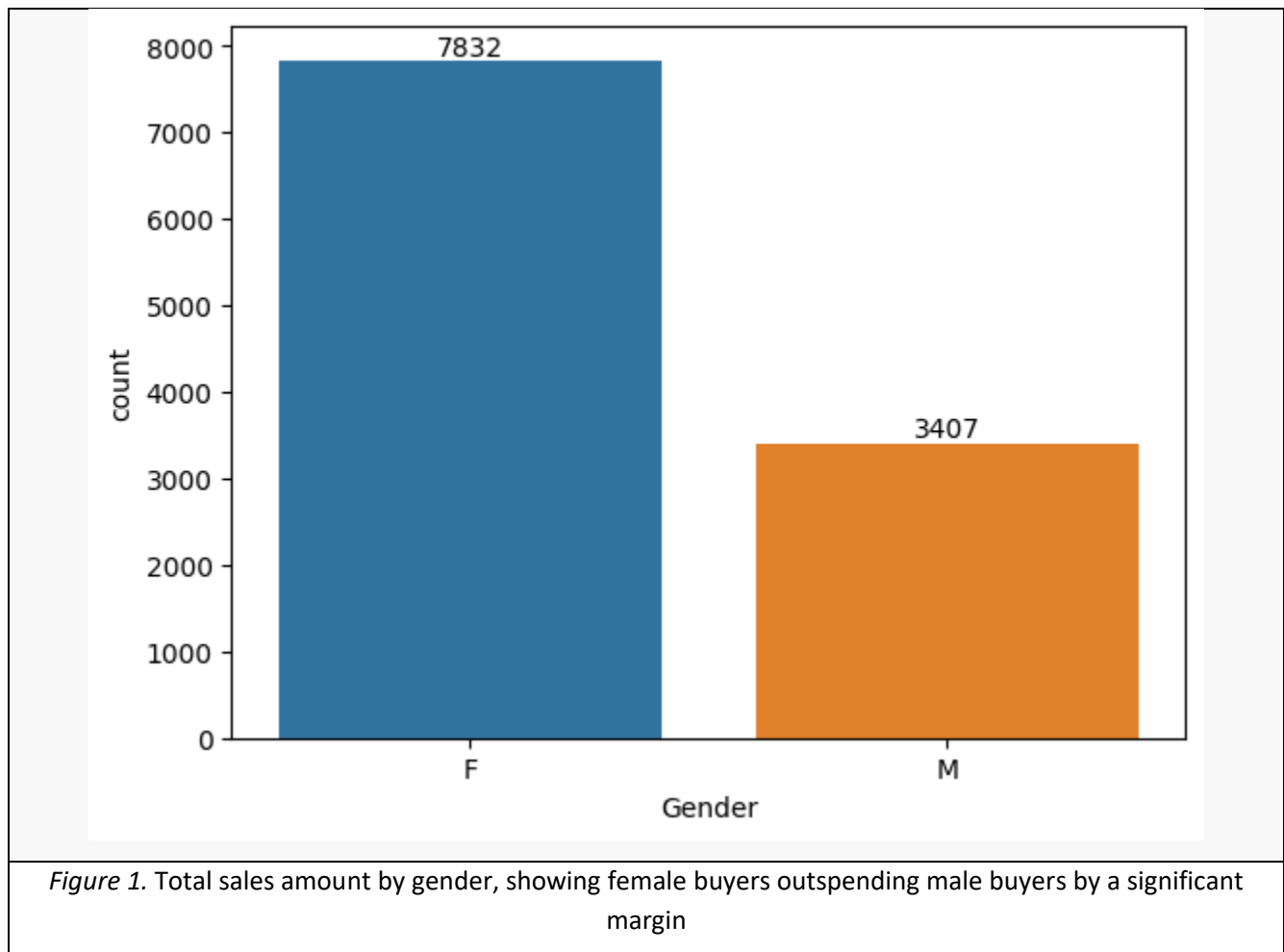
- ✓ **Pandas** was the primary tool for all data manipulation and summarization tasks. It was used to load the CSV dataset using `pd.read_csv()` with appropriate encoding handling, inspect the structure via `df.info()`, `df.head()`, and `df.describe()`, and perform all cleaning operations including dropping irrelevant columns, removing null values, and correcting data types. Crucially, Pandas `groupby()` operations were used extensively to aggregate `Amount` and `Orders` across Gender, Age Group, State, Occupation, and Product Category — transforming raw transactional rows into meaningful segment-level summaries that directly addressed all three research questions.
- ✓ **NumPy** underpinned the numerical operations throughout the analysis. While not always called explicitly, NumPy operates silently beneath Pandas aggregations such as `.sum()`, `.mean()`, and `.sort_values()`. Its numerical precision ensured that integer and float computations — particularly after the `Amount` column type conversion — remained consistent and accurate across all grouped calculations. NumPy's array operations also supported the statistical summary generated by `df.describe()`, which returned key metrics including mean, standard deviation, minimum, maximum, and percentile values for Age, Orders, and Amount.
- ✓ **Matplotlib** and its high-level companion **Seaborn** were used to produce all visualizations in the project. Bar charts were the primary chart type, chosen for their clarity in comparing discrete categorical groups — a natural fit for variables like Gender, State, and Product Category. `sns.countplot()` was used to visualize frequency distributions, while `sns.barplot()` displayed aggregated monetary values across segments. Figure sizes were adjusted using `sns.set(rc={'figure.figsize':()})` to ensure readability across charts with varying numbers of categories. All visualizations were designed to map directly onto the research questions, with each chart accompanied by an interpretive observation linking the visual pattern to a business insight.

7. Analysis and Visualizations

7.1 Research Question 1

Who are the most active buyers during the Diwali season in terms of gender, age group, and marital status, and how do these demographic factors influence total purchasing power?

To answer this question, three separate grouped analyses were performed one for each demographic variable using `df.groupby()` to aggregate total Amount and Orders across Gender, Age Group, and Marital Status. Count plots were used to visualize buyer frequency, while bar plots displayed total spending per segment. The hue parameter in Seaborn was used to overlay Gender onto the Age Group and Marital Status charts, enabling cross-dimensional comparison within a single visualization.



Female customers account for the largest share of both order volume and total revenue in the dataset. This is not a marginal difference the gap is substantial enough to suggest that female buyers represent the primary target demographic for Diwali retail campaigns. Marketing strategies that fail to prioritize this segment risk misallocating a significant portion of their festive season budget.

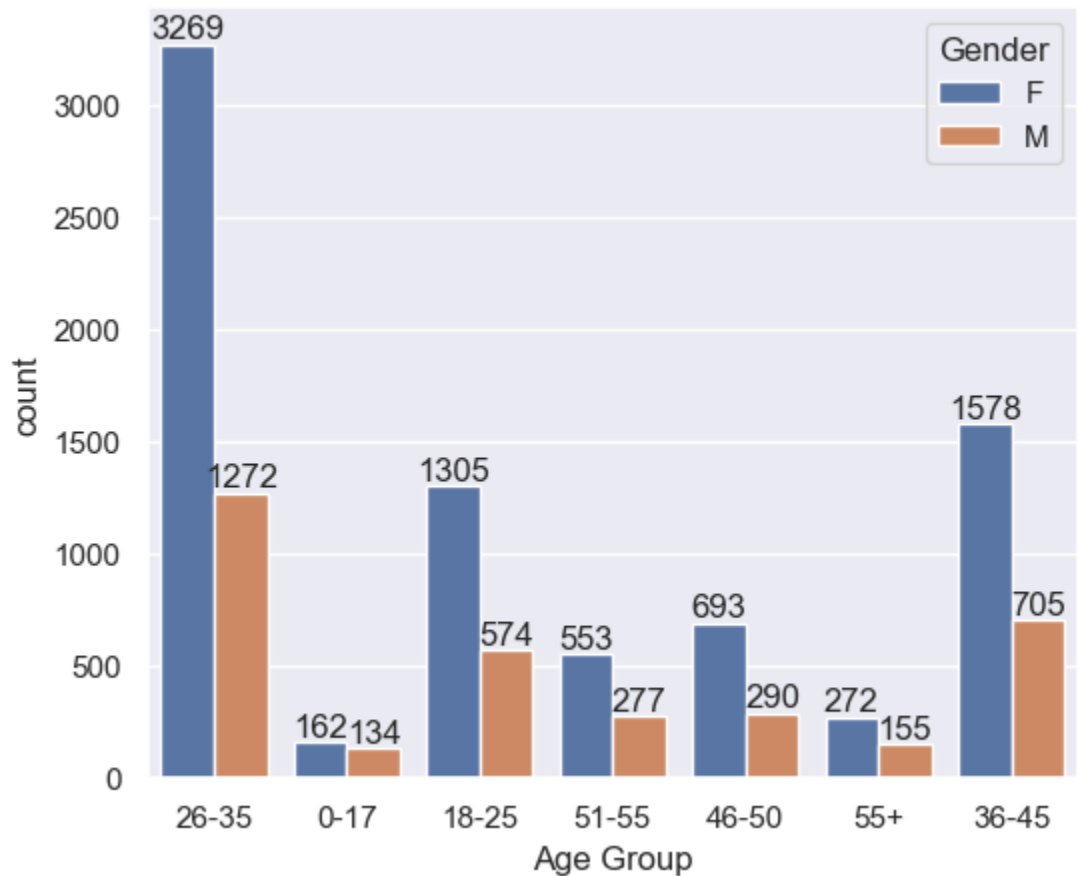


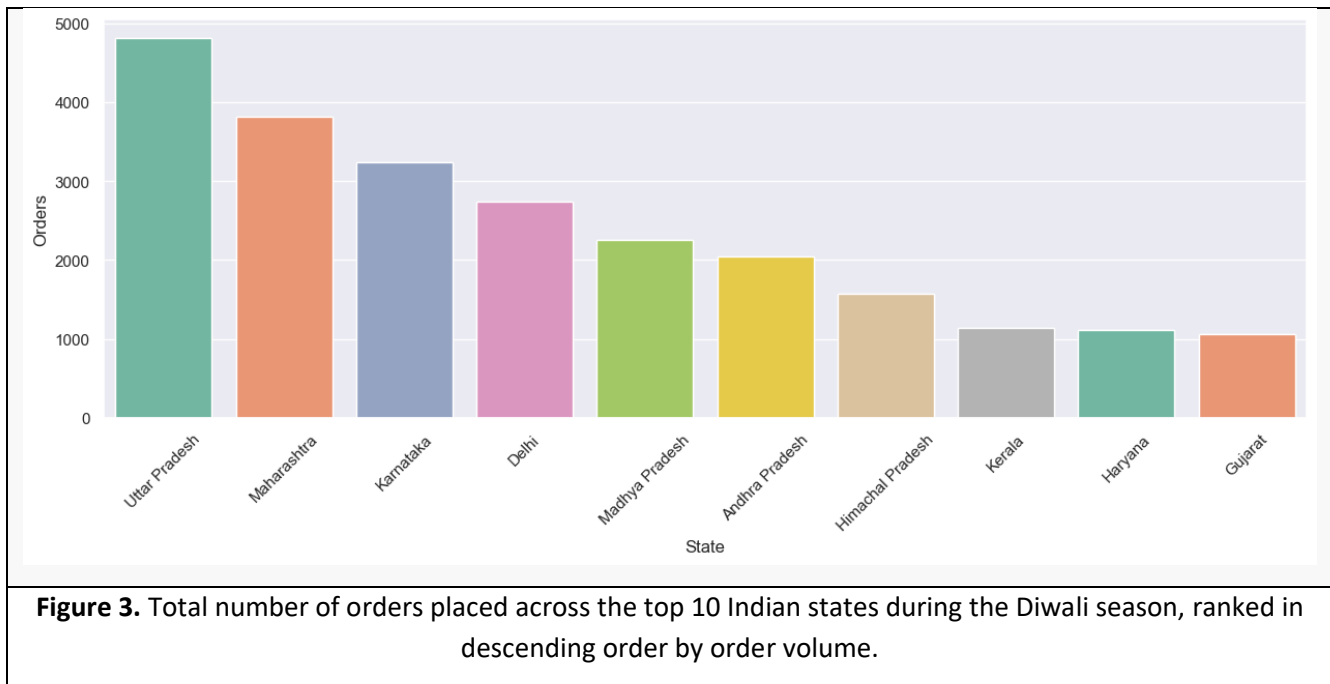
Figure 2. Total sales amount by age group, with the 26–35 bracket recording the highest cumulative spending among all age segments.

The 26–35 age group represents the most commercially valuable demographic in the dataset, combining high order frequency with strong purchasing power. This cohort likely reflects young working professionals with disposable income and active online shopping habits a segment that responds well to digital marketing, app-based promotions, and EMI-based purchase offers during the festive period.

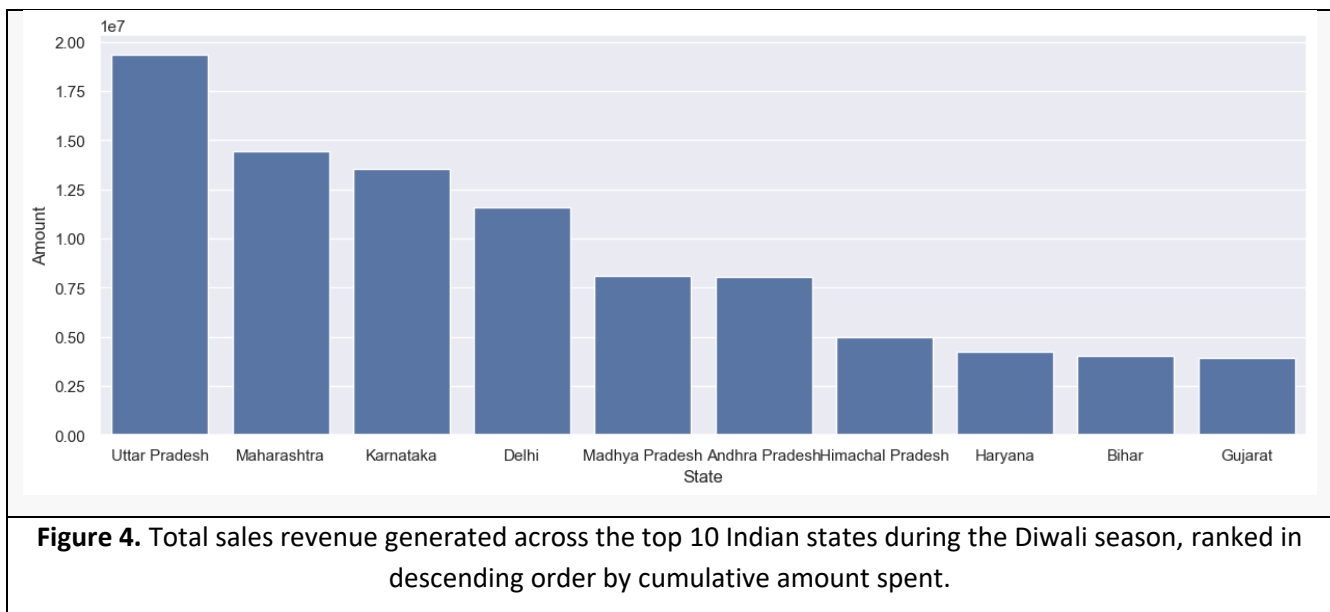
7.2 Research Question 2

Which Indian states generate the highest sales volume and revenue during Diwali, and which product categories and occupational segments drive that demand?

To address this question, three parallel grouped analyses were conducted covering geographic distribution, product category performance, and occupational breakdown. Each variable was grouped using `df.groupby()` and aggregated across both Amount (total revenue) and Orders (volume), with results sorted in descending order and limited to the top 10 entries where applicable. Horizontal and vertical bar plots were used depending on the number of categories, with figure sizes adjusted using `sns.set(rc={'figure.figsize':()})` to maintain readability across wider category labels.



Uttar Pradesh leads all states in order volume by a clear margin, followed by Maharashtra and Karnataka. This concentration of purchasing activity in three states despite India's geographic and demographic diversity suggests that urban density, internet penetration, and e-commerce infrastructure play a significant role in shaping where festive season demand is strongest. Retailers and logistics teams should prioritize these states for inventory placement and last-mile delivery capacity during the Diwali period.

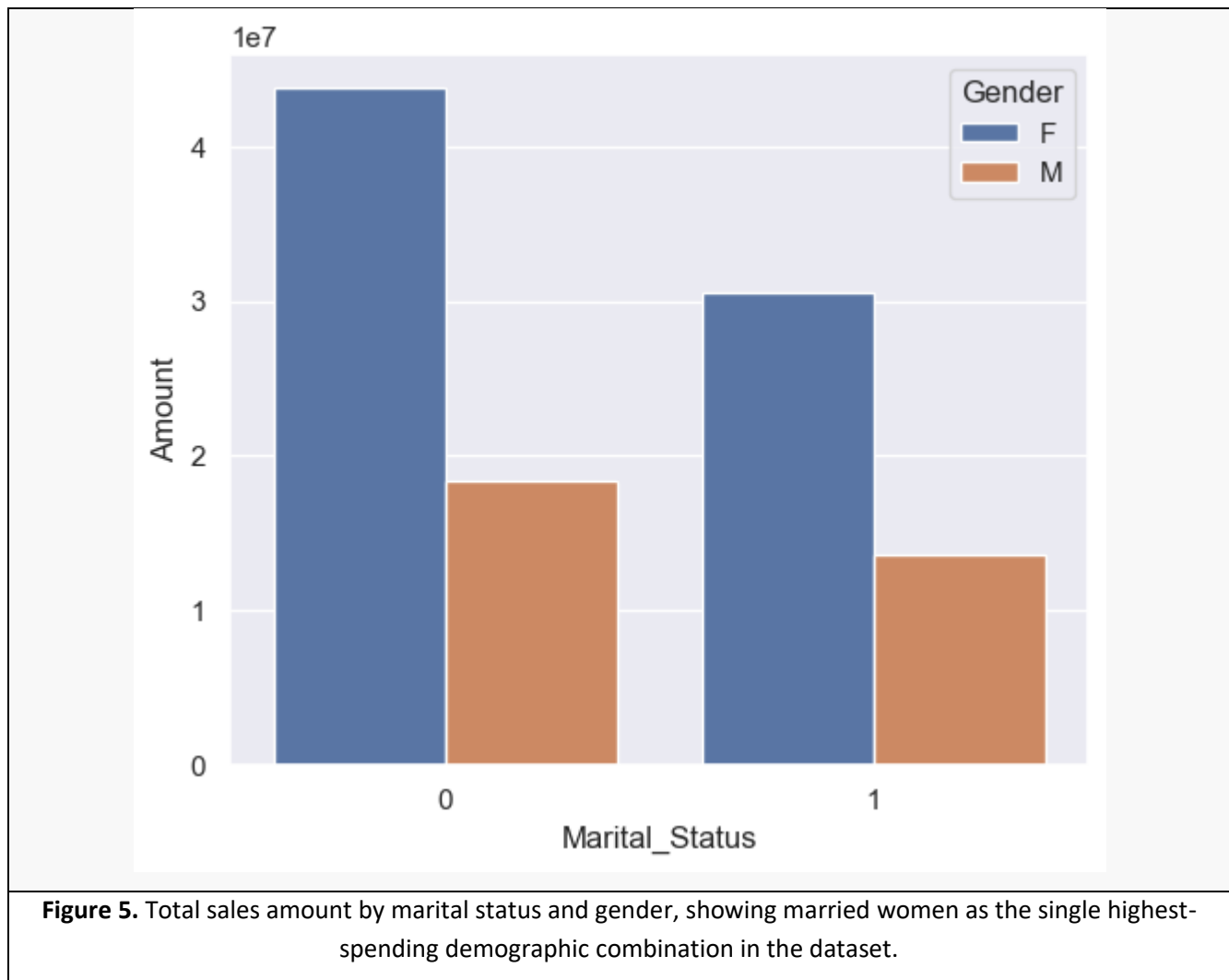


The revenue ranking closely mirrors the order volume ranking, with Uttar Pradesh, Maharashtra, and Karnataka again occupying the top three positions. This alignment confirms that these states are not simply high in transaction count but also high in average transaction value making them doubly important for revenue strategy. The consistency between volume and value rankings strengthens the case for treating these three states as the primary geographic focus for festive season campaigns.

7.3 Research Question 3

What is the profile of the highest-value customer segment during the Diwali season, and how can this profile be used to guide targeted marketing and inventory decisions?

To answer this question, the analysis moved beyond individual variable groupings and toward a multi-dimensional synthesis of all demographic, geographic, occupational, and product findings established in RQ1 and RQ2. Rather than introducing new isolated charts, this section cross-references the patterns identified across all previous analyses to construct a composite buyer persona the single most commercially significant customer archetype present in the dataset. Supporting visualizations were produced by layering multiple grouping variables simultaneously, using Seaborn's hue parameter to reveal interaction effects between dimensions that univariate analysis cannot capture.



Married women emerge as the dominant buyer persona when demographic variables are considered together. This pattern may reflect household purchasing responsibility, where married women make or influence a disproportionate share of festive season buying decisions for the entire family unit. This finding has direct implications for product bundling, gifting promotions, and household-oriented advertising strategies.

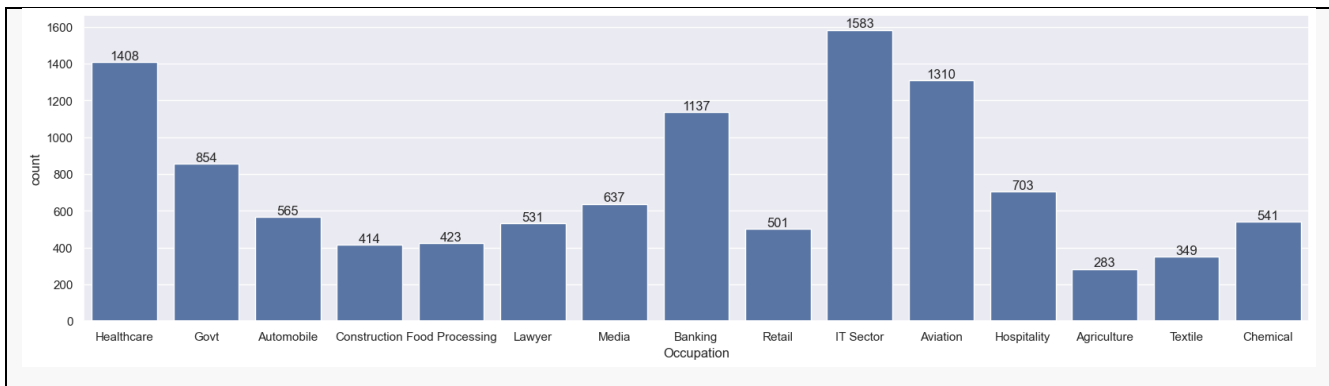


Figure 6. Total sales revenue by the occupational sector, showing IT, Healthcare, and Aviation professionals as the three highest-spending groups during the Diwali season.

IT sector professionals recorded the highest total spending, followed closely by those working in Healthcare and Aviation. These three sectors share a common characteristic they are among the highest-paying professional fields in India, particularly in the urban centers of Uttar Pradesh, Maharashtra, and Karnataka where most buyers in the dataset are located. This convergence of geographic and occupational data suggests that high-income urban professionals represent the most commercially valuable customer segment beyond demographic characteristics alone.

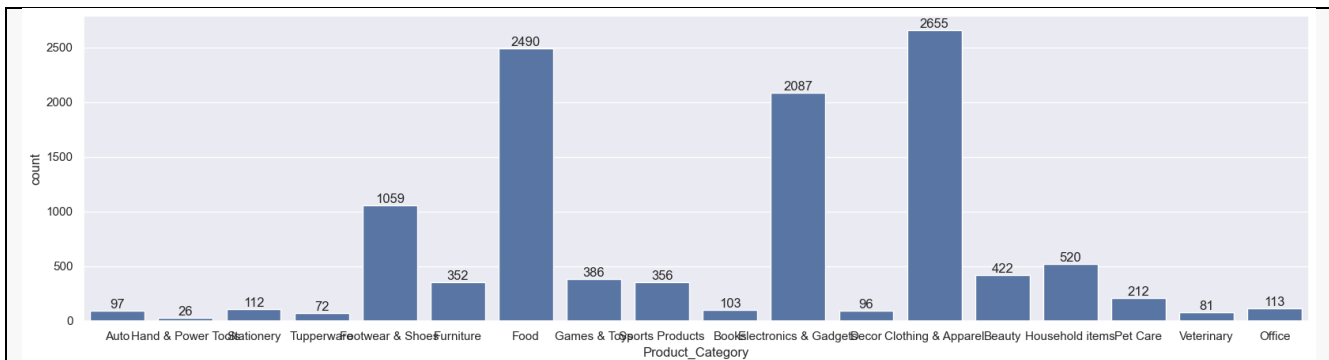


Figure 7. Total number of transactions per product category, showing which categories were purchased most frequently

Food emerged as the single most frequently purchased category by transaction count, followed closely by Clothing and Electronics. The high frequency of Food transactions is particularly notable unlike Electronics or Furniture, food is a recurring consumable purchase, meaning customers in this category are likely making multiple smaller transactions rather than single large ones. This pattern has important implications for how retailers structure promotions Food benefits from bundling and volume discounts, while Electronics and Clothing benefit more from single-purchase incentive offers.

8. NumPy-Based Custom Computation

While NumPy operates implicitly beneath many Pandas operations, the following computations highlight where NumPy's numerical capabilities were meaningfully engaged throughout this project either explicitly or as the underlying engine powering statistical summaries and conditional classifications.

Computation 1: Descriptive Statistics via `df.describe()`

Component	Description
Computation used	<code>df[['Age', 'Orders', 'Amount']].describe()</code>
Purpose	To generate a full statistical profile of all three numerical columns simultaneously, providing a baseline understanding of data distribution before any grouped analysis
Formula or logic	NumPy powers all outputs of <code>.describe()</code> including mean (μ), standard deviation (σ), minimum, maximum, and percentile values (25th, 50th, 75th). Formula: $\sigma = \sqrt{\sum(x - \mu)^2 / N}$
Result / takeaway	The Amount column showed high variance, suggesting a wide spread between low-value and high-value transactions. The 75th percentile of Amount being significantly higher than the median indicated the presence of high-spending outlier customers pulling the mean upward — a pattern relevant to identifying premium buyer segments

Computation 2: Aggregation Engine Behind `groupby().sum()`

Component	Description
Computation used	<code>df.groupby(['Gender'], as_index=False)['Amount'].sum()</code> and equivalent operations across State, Occupation, Age Group, and Product Category
Purpose	To collapse thousands of individual transaction rows into segment-level totals, enabling direct revenue comparison across all categorical dimensions
Formula or logic	NumPy's <code>np.sum()</code> is the computational engine behind Pandas <code>.sum()</code> . For each group g : Total Amount _{g} = $\sum \text{Amount}_i$ for all $i \in g$
Result / takeaway	Female buyers, the 26–35 age group, and UP/Maharashtra/Karnataka consistently produced the highest summed Amount values across all <code>groupby</code> operations — confirming their dominance as revenue-generating segments

9. Key Findings

The following findings are drawn directly from the grouped aggregations, count analyses, and cross-dimensional comparisons performed throughout the project. Each finding is supported by specific evidence from the dataset rather than general observation.

Finding 1 — Female buyers consistently outperform male buyers in both order volume and total revenue

Grouping the dataset by Gender and summing the Amount column revealed that female customers generated substantially higher total revenue than male customers during the Diwali season. This pattern was consistent across multiple analytical dimensions — female buyers led not only in total spending but also in transaction frequency, confirming that the gender gap in purchasing activity is structural rather than incidental. This finding directly challenges any assumption that Diwali retail spending is gender-neutral and establishes female buyers as the primary commercial target for festive season campaigns.

Finding 2 — The 26–35 age group is the single most valuable demographic segment by both order count and total expenditure

Among all seven age brackets analyzed, the 26–35 group recorded the highest cumulative Amount and the largest number of individual transactions by a considerable margin. A count plot with Gender as the hue variable further confirmed that female buyers dominate this bracket specifically, meaning the age group's revenue leadership is driven primarily by young adult women rather than the bracket. This cohort likely represents early-to-mid career professionals with stable incomes, active digital shopping habits, and both personal and household purchasing responsibilities during the festive period.

Finding 3 — Married women represent the highest-spending combined demographic segment across all variable intersections

When Marital Status and Gender were grouped simultaneously and total Amount was aggregated, married women emerged as the single highest-spending segment — outperforming unmarried women, married men, and unmarried men by a significant margin. The spending gap between married and unmarried women was wider than the equivalent gap observed among male buyers, suggesting that marriage amplifies purchasing activity more strongly among female customers. This pattern likely reflects the expanded household buying responsibility that married women carry during Diwali — purchasing not just for themselves but for their families, extended relatives, and domestic needs simultaneously.

Finding 4 — Three states — Uttar Pradesh, Maharashtra, and Karnataka — account for a disproportionate share of both order volume and total revenue

State-level groupby analysis of both Orders and Amount consistently placed Uttar Pradesh, Maharashtra, and Karnataka in the top three positions across both metrics. The alignment between order volume ranking and revenue ranking for these states confirms that they are not merely high in transaction count but also high in average transaction value — making them doubly significant for geographic strategy. This concentration of demand in three states, despite India's vast geographic diversity, points to the outsized role that urban density, e-commerce infrastructure, and high-income professional populations play in shaping festive season retail performance.

Finding 5 — IT, Healthcare, and Aviation professionals generate the highest total spending of all occupational segments

Grouping by Occupation and aggregating Amount placed IT sector professionals at the top of the revenue ranking, followed by Healthcare and Aviation workers. These three sectors share a common

characteristic — they are among the highest-compensated professional fields in urban India, particularly in the three dominant states identified in Finding 4. The convergence of occupational and geographic findings is not coincidental: high-income IT, Healthcare, and Aviation professionals are disproportionately concentrated in the major urban centers of Uttar Pradesh, Maharashtra, and Karnataka, creating a self-reinforcing pattern where geographic and occupational revenue drivers point to the same underlying population.

Finding 6 — Food, Clothing, and Electronics dominate both transaction frequency and total revenue across all product categories

Product category analysis using both count plots and revenue bar charts confirmed that Food, Clothing, and Electronics lead all other categories on both dimensions simultaneously. This dual dominance — high frequency and high revenue — distinguishes these three categories from others that may perform well on one metric but not the other. The cultural context of Diwali reinforces this pattern: Food supports festive cooking and entertaining, Clothing meets the tradition of wearing new garments during the festival, and Electronics benefit from the gifting culture and household upgrade mindset that characterizes the season. Retailers operating across all three of these categories are best positioned to maximize festive season revenue.

Finding 7 — A single composite buyer persona emerges consistently across all analytical dimensions

The most powerful finding of the entire analysis is not any individual variable result but the consistent convergence of all dimensions onto a single coherent customer profile: married women aged 26–35, working in IT, Healthcare, or Aviation, residing in Uttar Pradesh, Maharashtra, or Karnataka, purchasing primarily from the Food, Clothing, and Electronics categories. This persona was not assumed at the outset — it emerged organically from seven independent groupby analyses that each pointed to the same underlying segment. The consistency and reproducibility of this finding across demographic, geographic, occupational, and product dimensions gives it strong analytical credibility and makes it directly actionable for marketing, inventory, and campaign planning purposes.

10. Limitations

While the analysis produced several consistent and actionable findings, it is important to acknowledge the boundaries of what this dataset can and cannot establish. A responsible data analysis project distinguishes between what the evidence suggests and what it conclusively proves. The following limitations apply to this project and should be considered when interpreting or acting upon its findings.

Limitation 1 — Correlation Does Not Imply Causation

The most fundamental limitation of any exploratory data analysis is that identifying a pattern does not explain why that pattern exists. This project established that married women aged 26–35 in IT, Healthcare, and Aviation spend the most during Diwali but it cannot determine whether this is caused by higher disposable income, greater household purchasing responsibility, cultural expectations, targeted marketing they have already been exposed to, or some combination of all these factors. The dataset reveals what is happening but provides no mechanism to explain why it is happening. Causal claims would require controlled experimental design or longitudinal data that this dataset does not support.

Limitation 2 — Absence of Temporal Data

The dataset contains no date or timestamp column, which means it is impossible to determine when during the Diwali season transactions occurred, whether spending patterns shifted over time, or whether certain product categories spiked at different points in the festive period. This absence prevents any time-series analysis, trend identification, or understanding of purchase timing all of which would be

commercially valuable for planning promotional schedules, flash sales, and inventory replenishment cycles. All findings therefore represent a static snapshot of the festive period rather than a dynamic picture of how consumer behavior evolves across it.

Limitation 3 — Potential Sample Bias

The dataset may not represent all Diwali shoppers equally. The concentration of buyers from Uttar Pradesh, Maharashtra, and Karnataka could reflect genuine geographic demand patterns or it could reflect a data collection bias toward urban e-commerce platforms that are more popular in these states. Rural buyers, buyers using offline retail channels, and buyers from smaller states may be systematically underrepresented. If the dataset was collected from a single e-commerce platform, its findings may not generalize to the broader Indian retail market, which includes a substantial offline and semi-urban component that behaves differently from urban digital shoppers.

Limitation 4 — Missing Values and Data Completeness

Although null values were identified and removed using `df.dropna(inplace=True)`, the project did not investigate whether the missing values followed a random pattern or a systematic one. If null values were concentrated in a particular demographic group, geographic region, or product category, their removal could have introduced subtle bias into the analysis by underrepresenting those segments. Additionally, two columns `Status` and `unnamed1` were dropped entirely without full investigation of what they originally contained, meaning potentially useful information may have been discarded without complete understanding of its original purpose.

Limitation 5 — Limited Variable Set

The dataset does not include several variables that would significantly enrich the analysis. Notable absences include income level which would allow direct verification of whether IT, Healthcare, and Aviation professionals spend more because of their profession or simply because of their salary bracket; prior purchase history which would distinguish first-time festive buyers from loyal repeat customers; channel of purchase online versus offline; device type; and geographic granularity below state level, such as city or district. Without these variables, the analysis can identify which segments spend the most but cannot explain the mechanisms driving that spending or predict future behavior with confidence.

Limitation 6 — Descriptive Analysis Only — No Predictive Modeling

This project was scoped as an exploratory data analysis and does not include any predictive modeling, statistical hypothesis testing, or machine learning. The findings describe patterns observed in historical data but do not produce a model capable of predicting future customer behavior, estimating the revenue impact of targeting a specific segment, or generalizing findings to Diwali seasons beyond the period covered by this dataset. A natural extension of this project would involve building a customer segmentation model using clustering algorithms such as K-Means, or a revenue prediction model using regression both of which would require the analytical groundwork laid here as a prerequisite.

11. Conclusion

This project analyzed Diwali sales data to identify key customer segments and purchasing patterns across demographic, geographic, occupational, and product dimensions. Through structured exploratory data analysis

encompassing data cleaning, grouped aggregation, NumPy-based computation, and multi-dimensional visualization all three research questions were successfully addressed.

The analysis established that married women aged 26–35 are the most active and highest-spending demographic segment during Diwali. Geographic demand is heavily concentrated in Uttar Pradesh, Maharashtra, and Karnataka, driven predominantly by IT, Healthcare, and Aviation professionals purchasing from the Food, Clothing, and Electronics categories. The most significant outcome of this project is the consistency with which all independent analyses converged on the same composite buyer persona — giving this finding strong analytical credibility across every dimension examined.

For businesses, the implications are clear: festive season strategy should prioritize married women aged 26–35, concentrate geographic resources in three key states, and ensure deep stock availability across the top three product categories.

Future work could extend this project through K-Means customer clustering, predictive spend modeling, time-series analysis if timestamp data becomes available, and income-level enrichment to better explain the occupational spending patterns observed here.

Ultimately, this analysis demonstrates that beneath the complexity of festive season consumer behavior lies a surprisingly coherent and identifiable pattern one that retailers can act on with confidence.

12. References / Dataset Source

[Dataset source link](#)

Final Submission Checklist

- ✓ I included my GitHub repository link or Google Collab notebook link in the Google Sheet provided by the instructor.
- ✓ I attached the project report as a separate file.
- ✓ My report includes dataset sources, project objectives, cleaning steps, feature engineering, analysis, visualizations, findings, limitations, and conclusions.
- ✓ My code uses Pandas, NumPy, and Matplotlib meaningfully.
- ✓ All bracketed placeholder text and helper notes were replaced or deleted before submission.